GitHub Copilot Customization Architecture

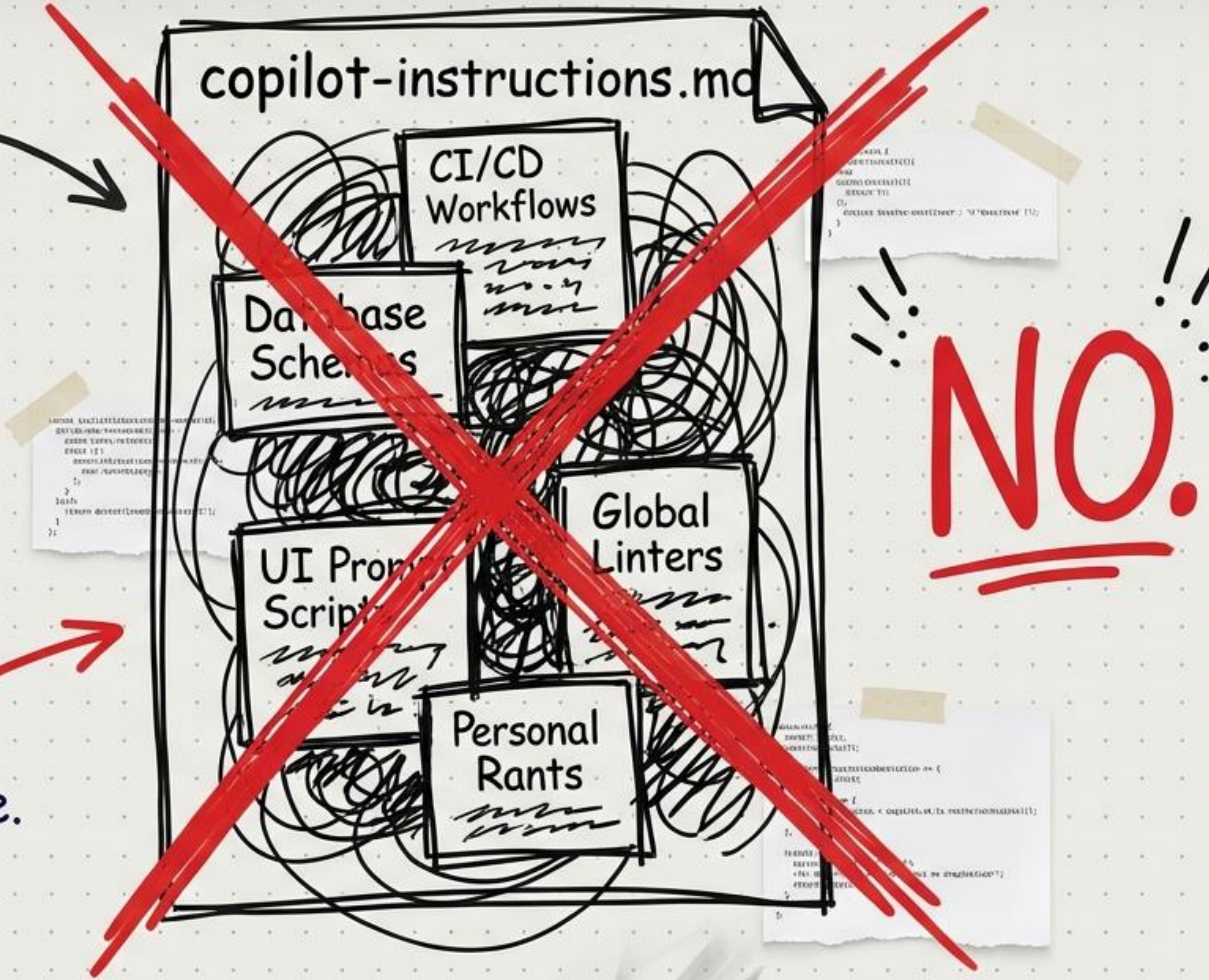
Instructions · Skills · Agents · Hooks · SDK

V2 — Please don't blow the token budget this time.
— Management.

The 2025 Anti-Pattern:

Shoving the entire engineering org into one always-on markdown file.

- It wastes tokens.
- It confuses the LLM.
- It destroys performance.



The Clean Slate: A 5-Layer Cooperative Architecture.

Foundation	Library	Org Chart	Safety	App
Instructions	Agent Skills	Custom Agents	Hooks	Copilot SDK
"Who you are" (Standards & Bylaws)	"How to do it" (Specialist Manuals)	"Who does it" (Dedicated Roles)	"The Bouncers" (Checkpoints)	"Remote Control" (Programmatic)

DO NOT ERASE

Step 1: The Instruction Bedrock

Copilot merges all active layers simultaneously to form the core context.

Highest Priority

1. Personal (VS Code Settings)
2. Repository-Wide (.github/copilot-instructions.md)
3. Path-Specific (*.instructions.md via applyTo glob)
4. Agent Instructions (AGENTS.md)
5. Model-Specific (CLAUDE.md / GEMINI.md)
6. Organization (GitHub Org Settings)

Base Layer

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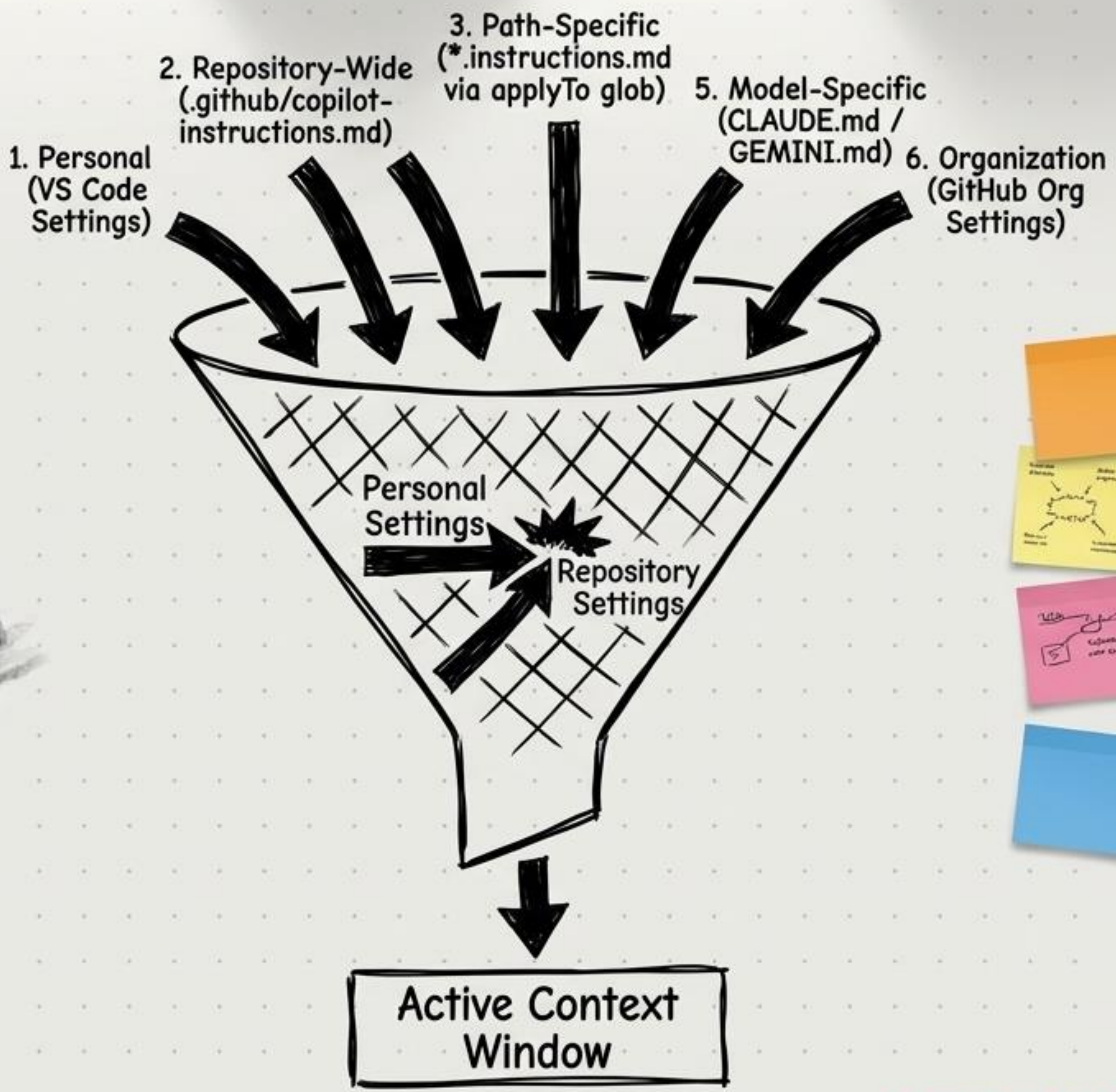
Precedence Rules:

- The closest layer to the user always wins.
- Layer 0 (Personal) overrides Layer 1 (Repo).
- Avoid contradictions acnnoat across layers.



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Instructions are standards, NOT runbooks.

- Keep repo-wide instructions to 5–15 core rules (200–500 tokens).

- Use them for things that apply to EVERY single interaction

- Use them for things that apply to EVERY (e.g., 'Use TypeScript strict mode').

If you put your DB migration script in here, it reads it every time you ask it to center a div. Stop doing that.

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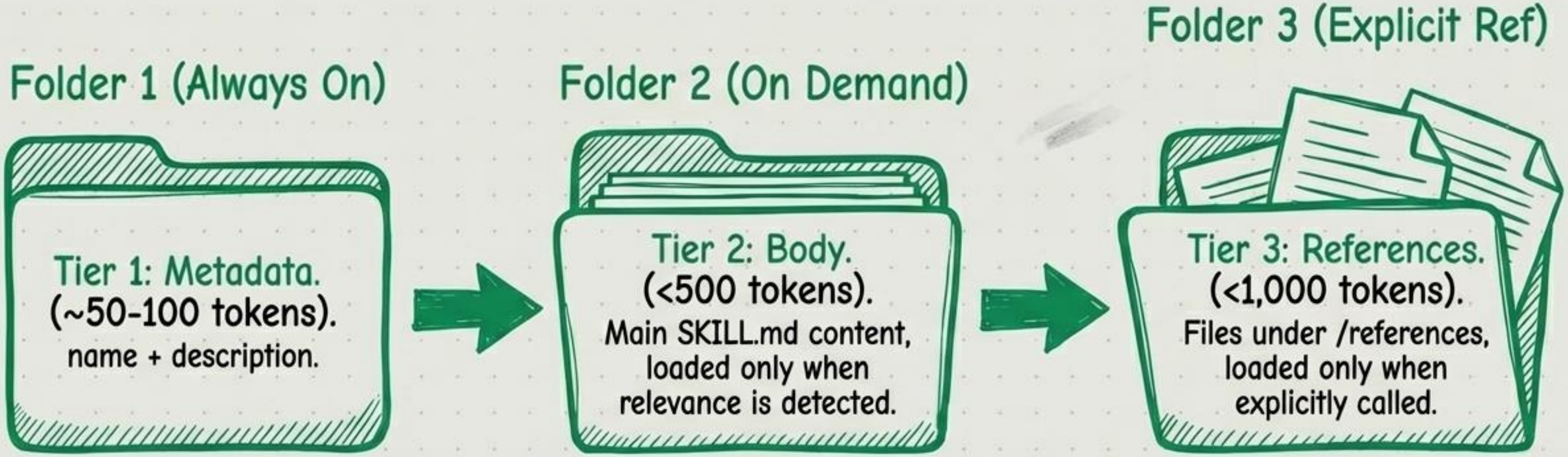
Step 2: Progressive Disclosure (Agent Skills)

Only pay for the tokens you actually need.



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```
---
name: webapp-testing

description: Toolkit
for testing local web
apps using Playwright.
Use when asked to
verify frontend
functionality or UI
behavior.
---

## Quick Start...
```

Copilot's ONLY way to know this skill exists.

Be descriptive, or it gathers dust on the shelf.

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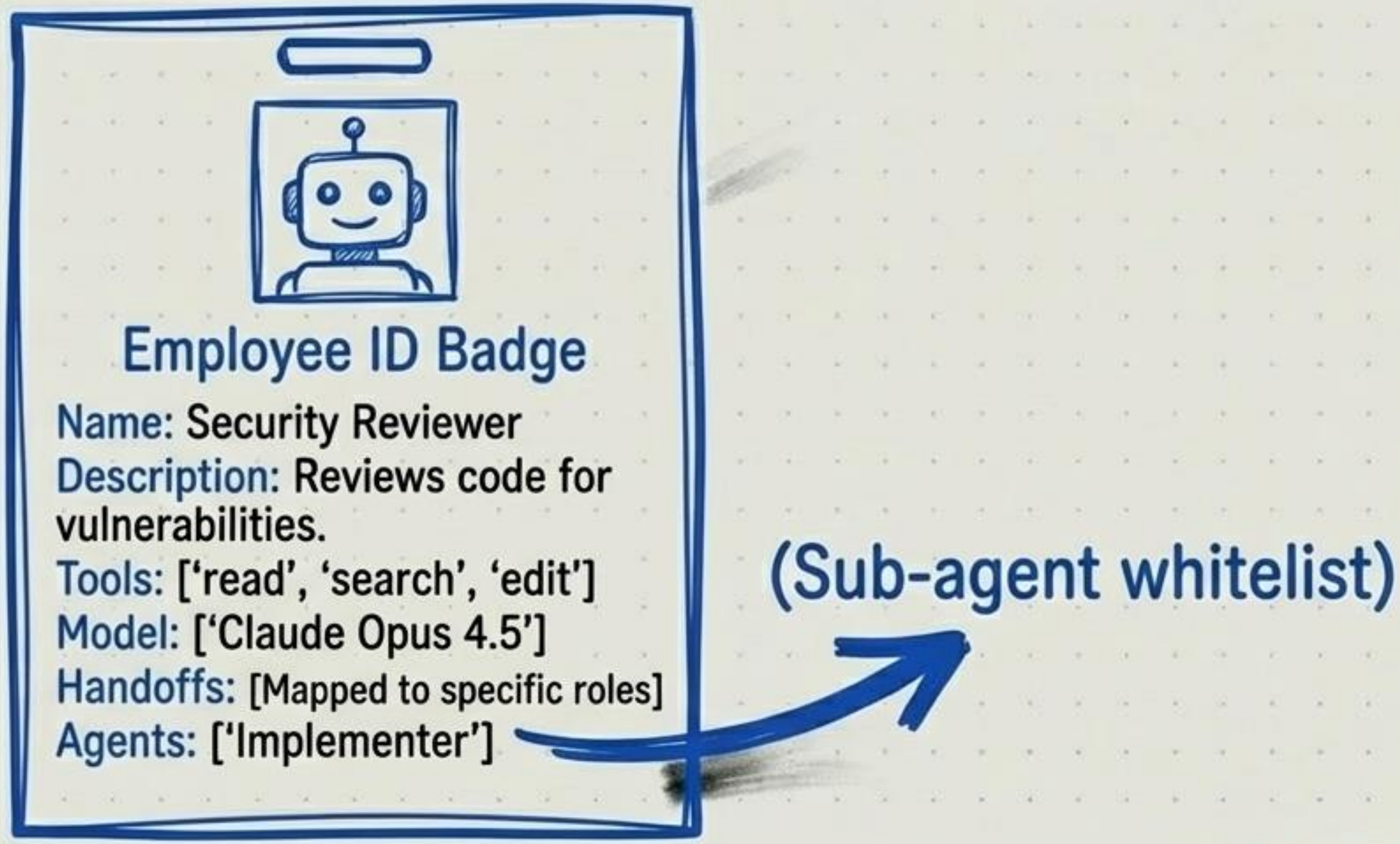
Step 3: Defining the Workforce (.agent.md)

Every file establishes a strict role, persona, and capability bound for a Custom Agent.



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Choose Your Delegation Strategy: Handoffs vs. Sub-agents

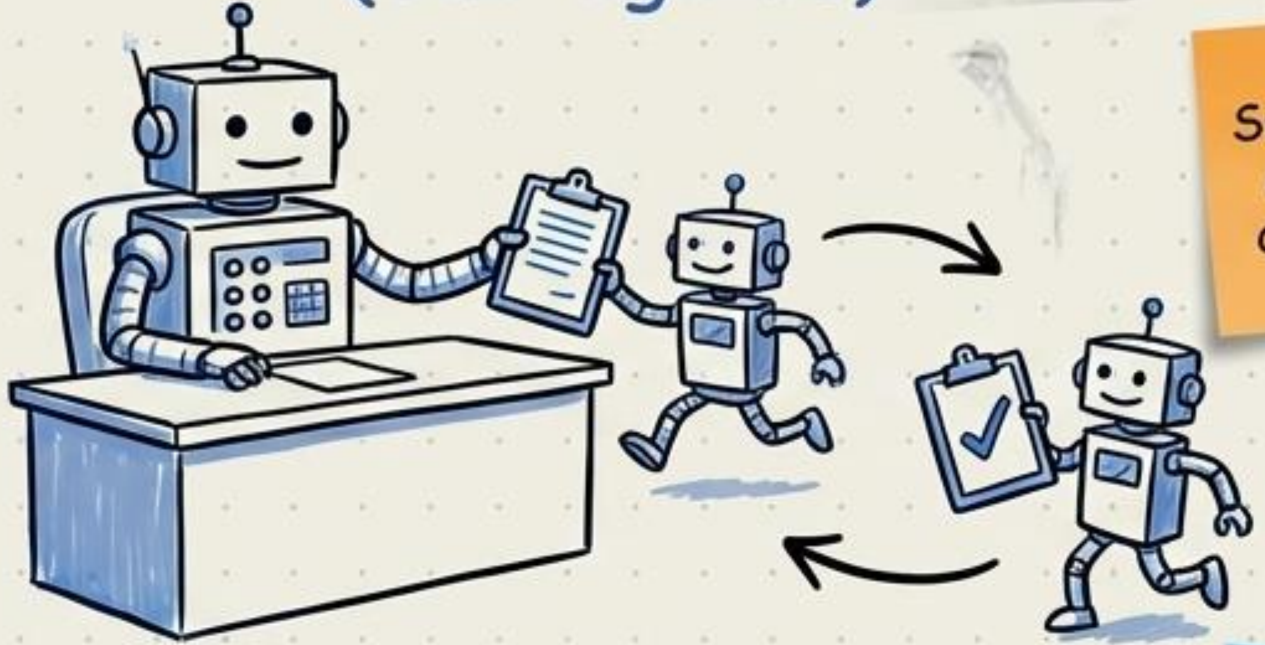
Remember: Context transfer is key!

Check token costs for deep pipelines

The Relay Race (Handoffs)

- User-driven trigger
- Shared conversation history
- Multi-stage pipelines
- Unlimited depth limits

The Manager Desk (Sub-agents)



Sub-agents are self-contained.

- Autonomous agent trigger
- Isolated context window
- Divide-and-conquer tasks
- Single-level depth limit

Manager handles coordination.

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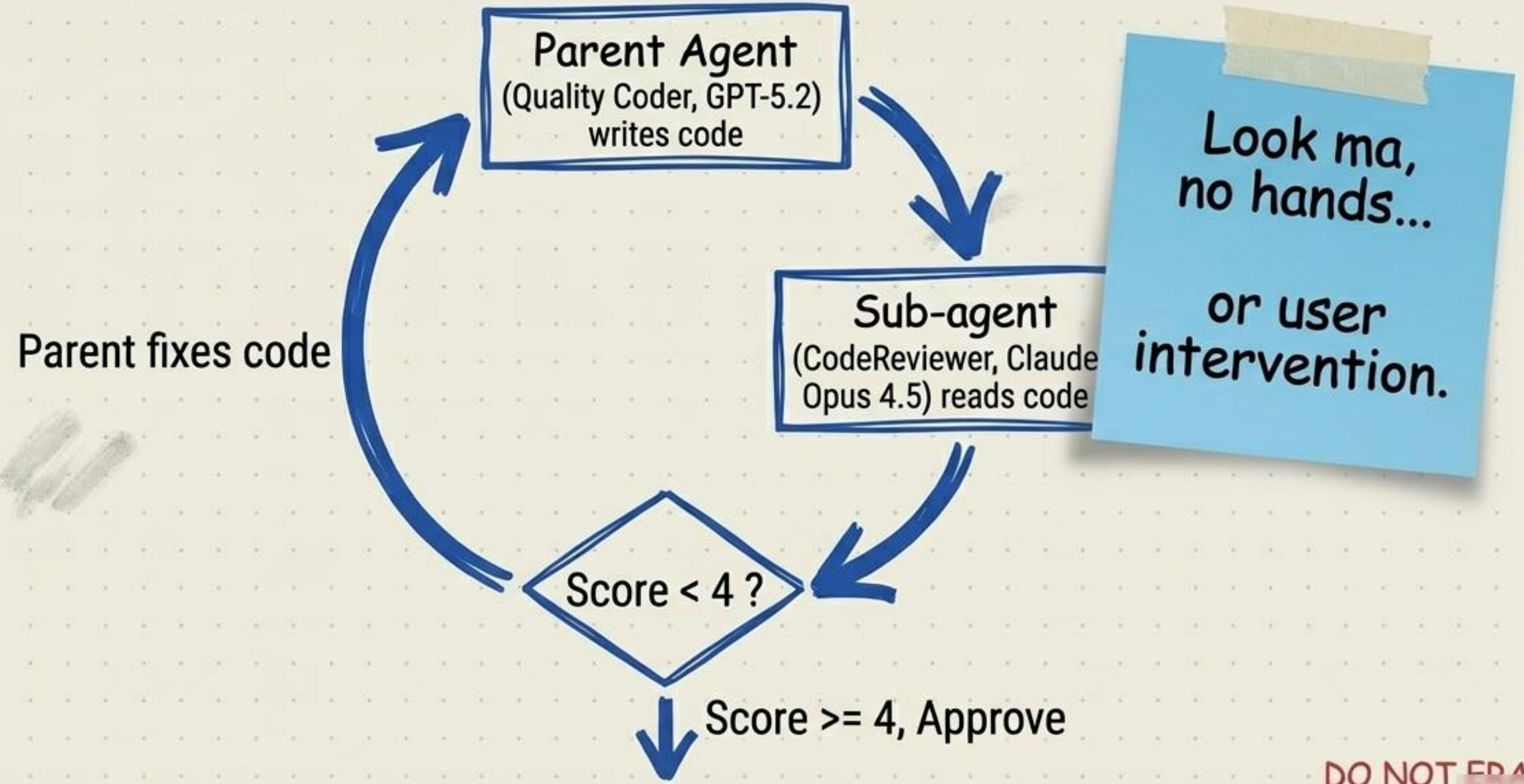
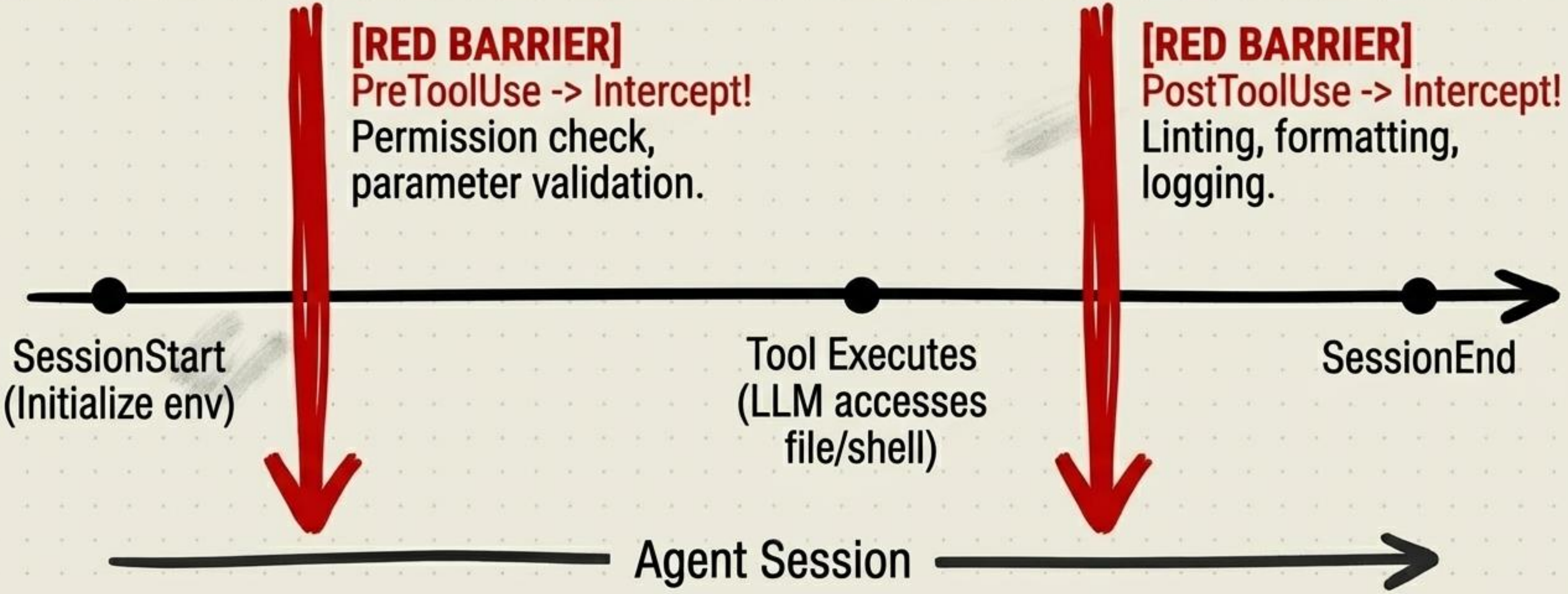
The Automated Review Loop

Heterogeneous models evaluating each other in isolated contexts.



Step 4: The Bouncers (Hooks)

Deterministic code executing at strict lifecycle checkpoints.



Zero Tokens.

The CFO's favorite feature. Put your linting and basic checks here, not in the LLM prompt.

Hooks are standard deterministic scripts. They execute at the system level and bypass the LLM entirely.

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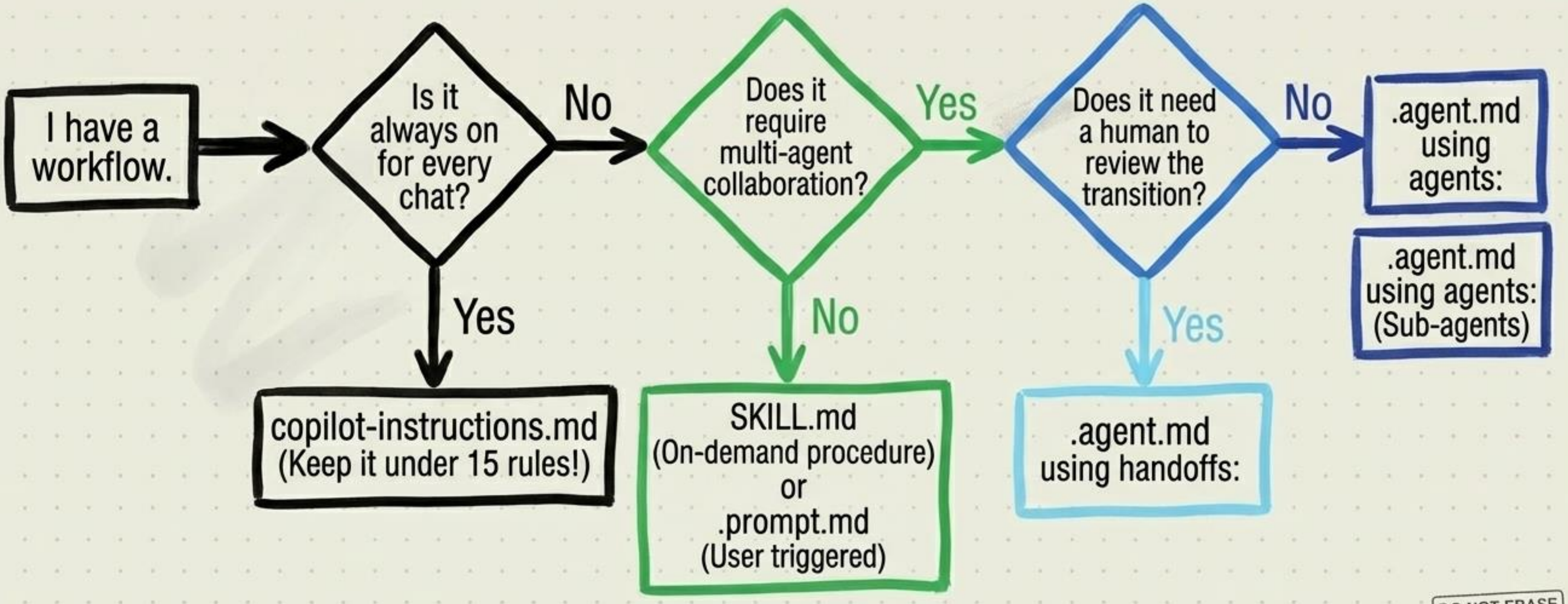
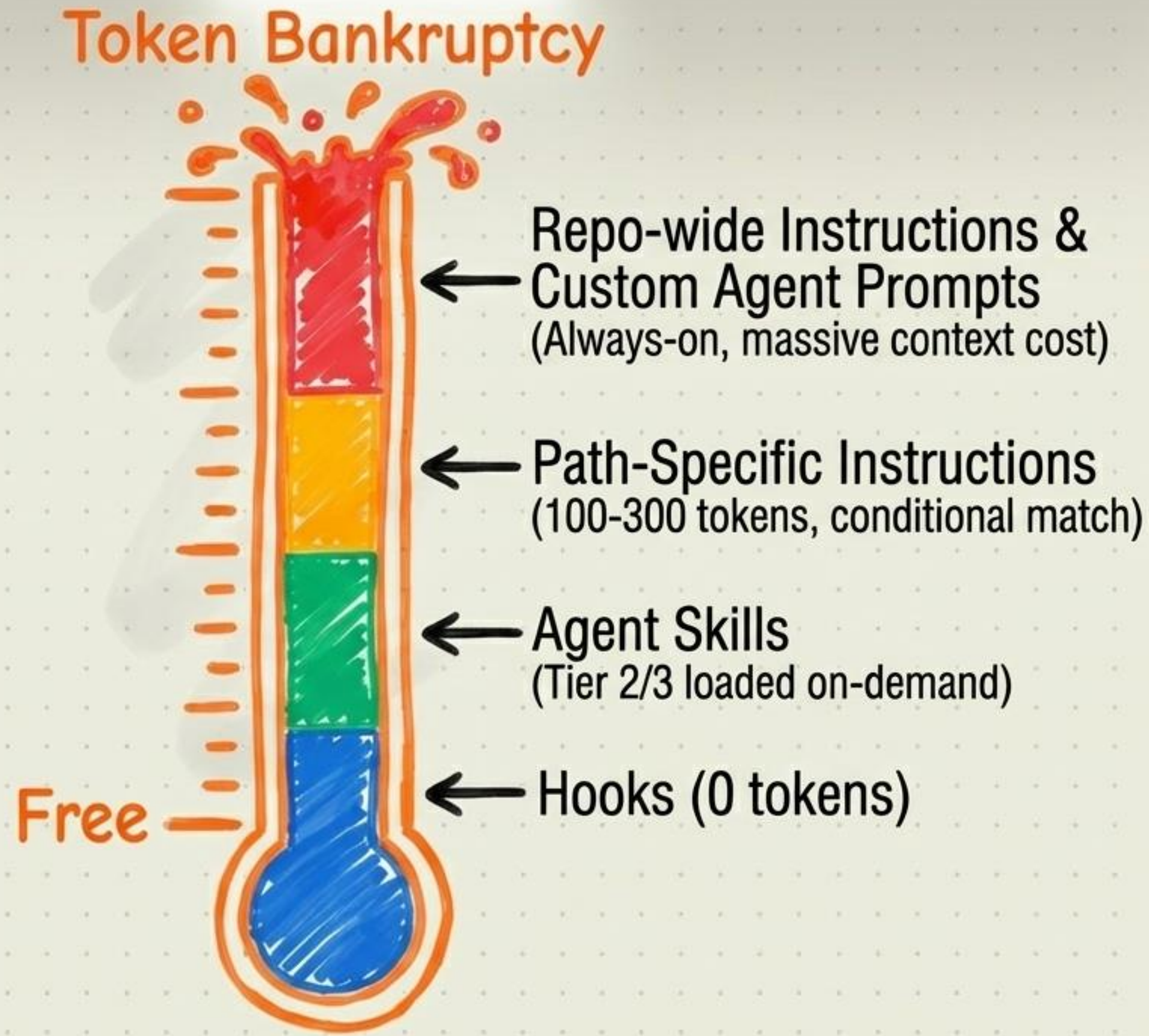
Where Does the Workflow Belong?

Stop putting orchestration logic in your global instructions.



The Token Thermometer of Pain.



- Never always-on what can be loaded on-demand.
- Never prompt what can be hooked.

- Default Deny: Agents cannot execute shell or read/write without explicit allowlists.
- Least Privilege: Planners get “search”. Implementers get “edit”. Do not mix them.
- Container Isolation: Shell access belongs in Dev Containers.
- Audit Trails: Use PostToolUse hooks to log every single invocation.

Do NOT give the planning agent `rm -rf’ permissions. Just don’t.

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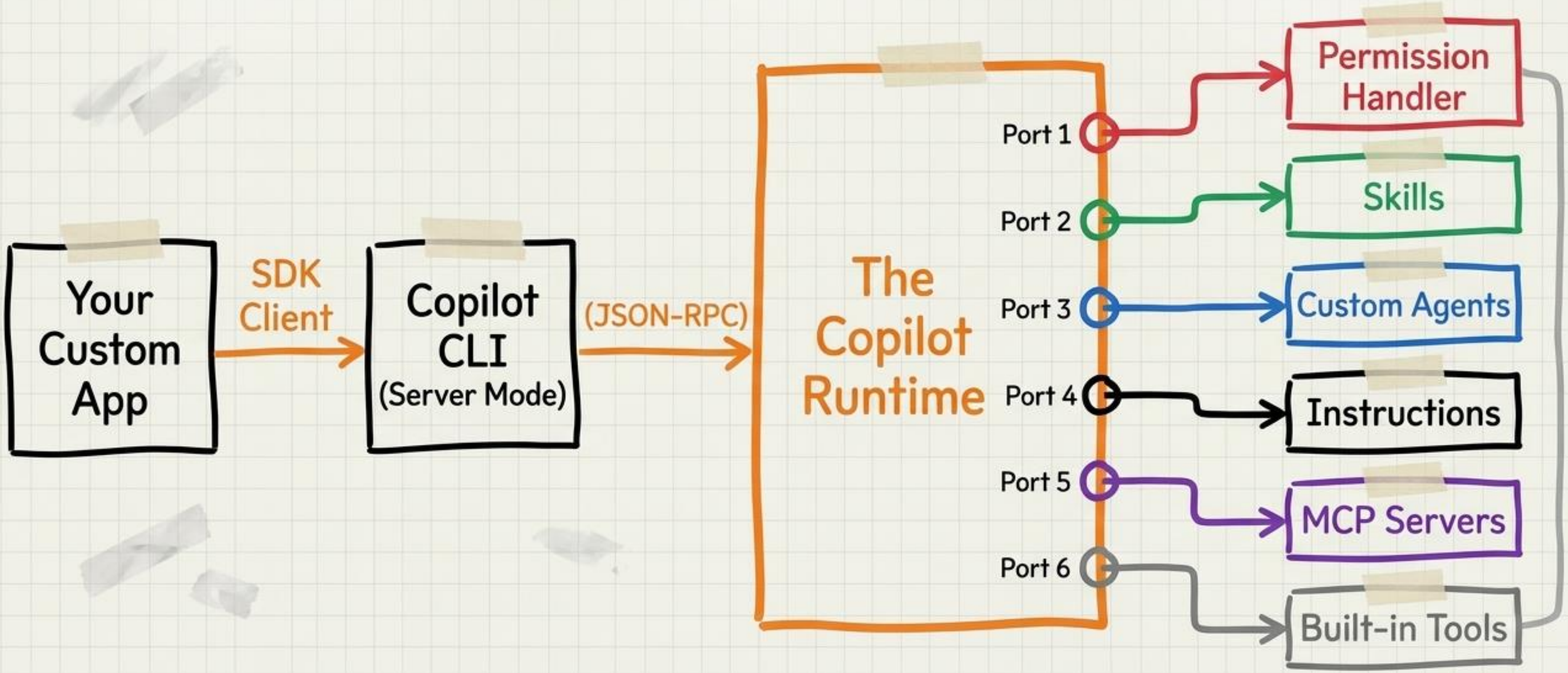
Step 5: The Remote Control (SDK)

Assembling the architecture into an invocable programmatic runtime.



The Permission Handler.

The SDK's built-in PreToolUse hook for default-deny operations.

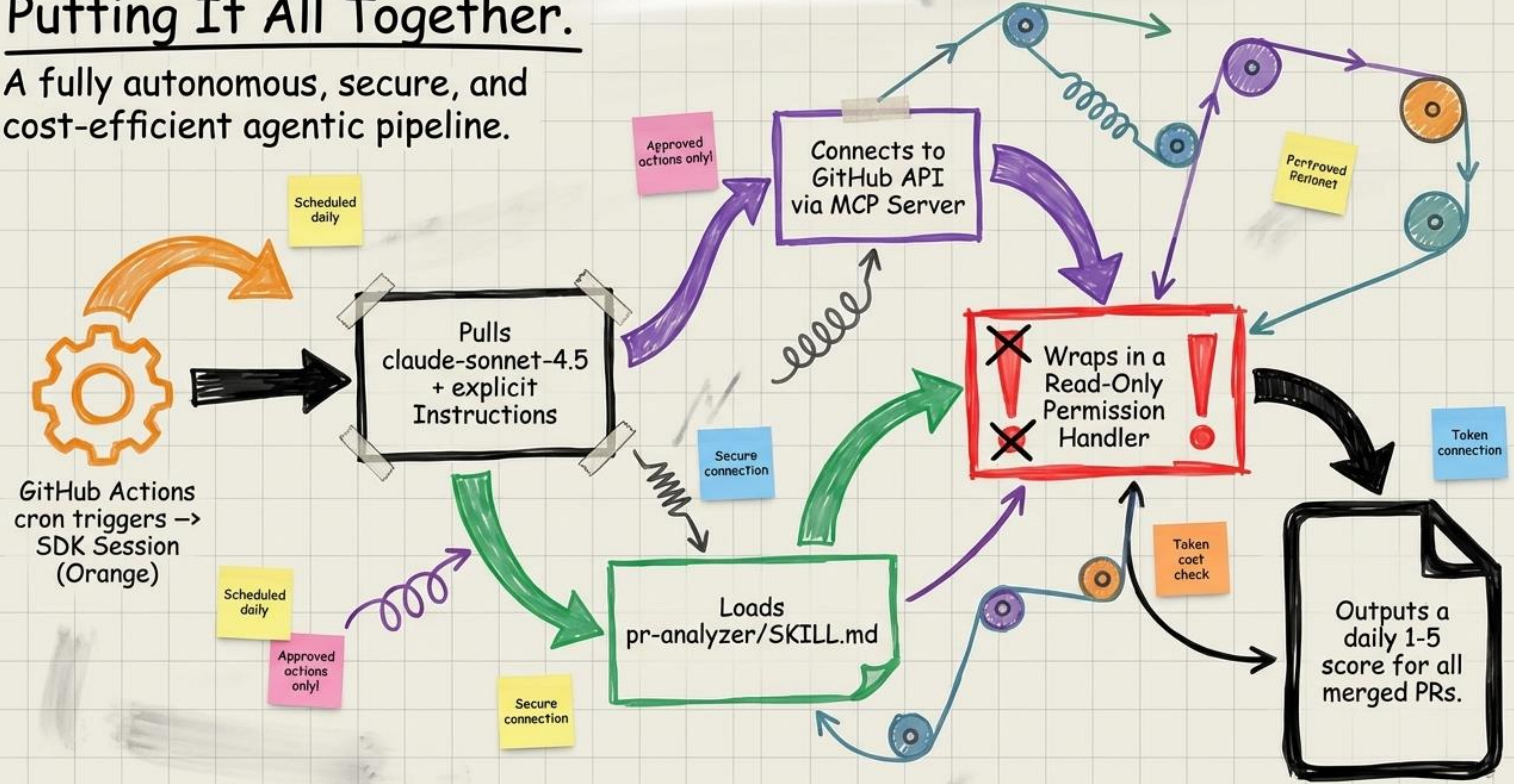
```python
def permission_handler(request, context):
    cmd = context.get("command", "")

    if any(cmd.startswith(prefix) for
    prefix in ["npm test", "pytest"]):
        return {"kind": "approved"}

    if "rm -rf" in cmd or "sudo" in cmd:
        return {"kind": "denied"}
```

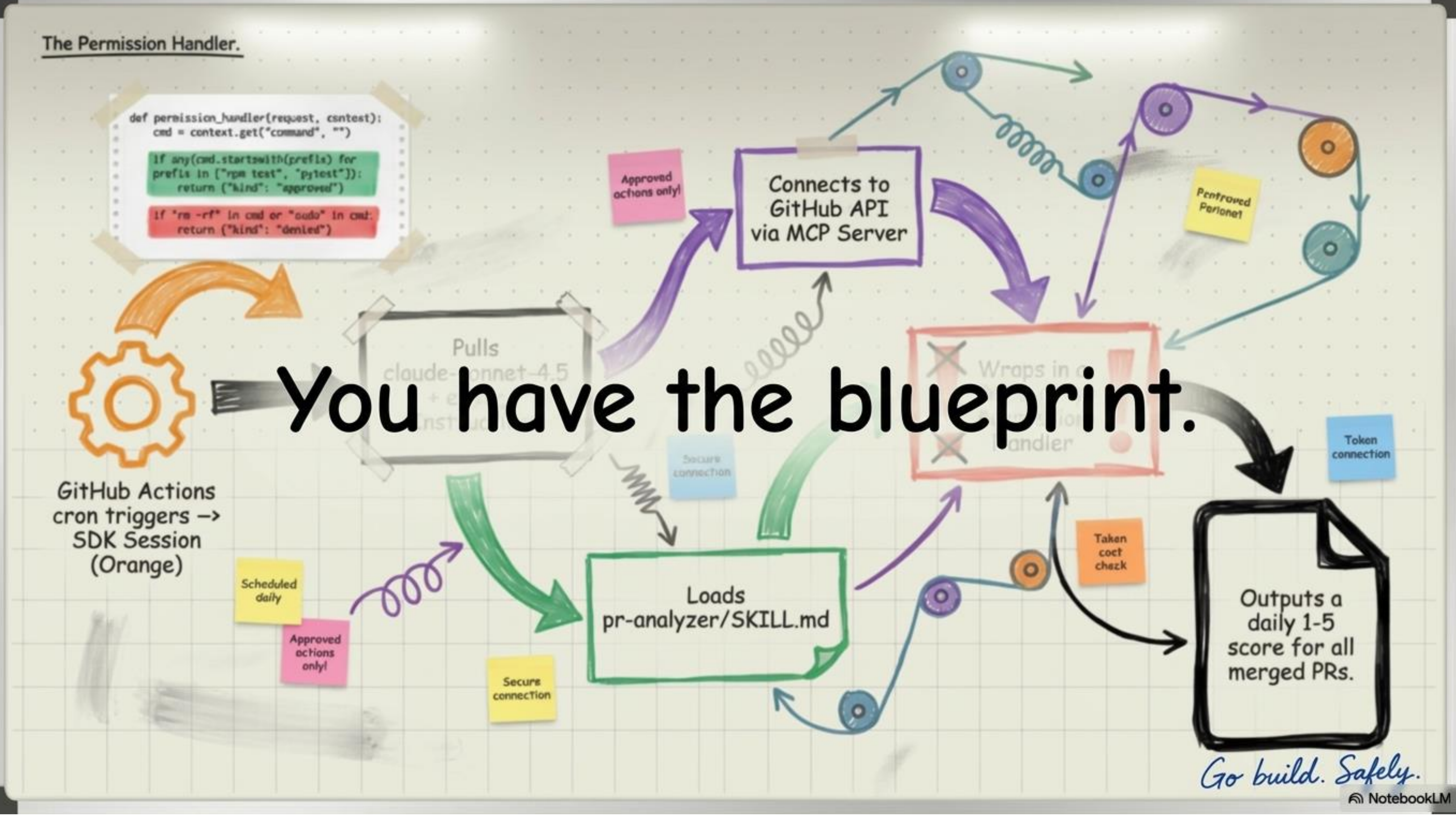
Allowlist Strategy

Blocklist Strategy

Putting It All Together.

A fully autonomous, secure, and cost-efficient agentic pipeline.





The Permission Handler.
def permission_handler(request, csntest):
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prefix in ["npm test", "pytest"]):
return ("kind": "approved")
if "rm -rf" in cmd or "sudo" in cmd:
return ("kind": "denied")
Approved actions only!
Connects to GitHub API via MCP Server
Pentroved Perionet
Pulls
You have the blueprint.
Token connection
GitHub Actions cron triggers -> SDK Session (Orange)
Secure connection
Scheduled daily
Approved actions only!
Loads pr-analyzer/SKILL.md
Taken coct check
Outputs a daily 1-5 score for all merged PRs.
Secure connection
Go build. Safely.
NotebookLM